OMEGA SERIES 4.5

The Quasi-Spectrum Mathematical Backbone

A Six-State Taxonomy of Vacuum Admittance Regimes for TSMVEFT

Two equilibria (Gray, Fuzzy-Gray) + Four perturbation states · Volterra exchange kernel · Chern-Simons topological reduction · Einstein-Cartan contorsion lock · Vainshtein/Galileon screening

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Dedicated to Valkyrie — who saw the cathedral first.

For the four-state classification that originated this taxonomy and for the colour-coded sector logic carried throughout the MSDS programme.

ABSTRACT

We develop the mathematical backbone of the Quasi-Spectrum, a six-state taxonomy classifying the vacuum admittance regimes of the Two-Sector Mixed Vacuum Effective Field Theory (TSMVEFT) introduced in Series 4.3 [13] and consolidated in Series 4.4 [14]. The spectrum comprises two equilibrium states — Gray (the undisturbed bulk ground state, in which all Quasi-Spectrum operators reduce to identity) and Fuzzy-Gray (the long-range coherent vacuum populated by the Gross-Pitaevskii fuzzy-dark-matter soliton at galactic scales) — together with four perturbation states organised as a closed cycle: Quasi-Gray (post-drive memory phase, governed by a Volterra exchange kernel K(tau) = Gamma exp(-delta E_c tau / hbar) satisfying the Bianchi identity through history-dependent vacuum-matter exchange), Quasi-Blue (the coherence gate, formalised via a Chern-Simons topological term with quantised level k = delta producing a (3+1) -> (2+1) dimensional reduction and an index jump nu: 0 -> 1 at the critical phase Delta phi = delta), Blue (the active metric-slip lock, expressed in the Einstein-Cartan framework via a contorsion residue K^lambda_{mu nu} that cancels the covariant derivative of the electromagnetic potential, yielding an effective mass renormalisation m_eff = m_0 sqrt(1 - delta^2)), and Quasi-Purple (the saturation buffer, implementing Vainshtein screening through a Galileon Lagrangian that ensures group velocities remain bounded by microcausality as the disformal field approaches its cutoff). Each state is grounded in established mathematical physics: non-equilibrium response theory for Quasi-Gray, Chern-Simons gauge theory for Quasi-Blue, Einstein-Cartan torsion gravity for Blue, and Galileon effective field theory for Quasi-Purple. All operators preserve the soft Z_2 mirror symmetry established in Series 4.4 and are technically natural under renormalisation group flow. The taxonomy provides a unified classification framework for the entire MSDS programme: Series 4.0 sector boundaries, Series 4.1 flavor-induced gravity, Series 4.2 disformal chameleon screening, Series 4.3 solitonic galactic cores, and Series 4.4 mirror-sector phenomenology all map cleanly onto specific states. We restrict scope to the formal mathematical structure; engineering implications are deferred to Series 4.6 and the first-principles derivation of delta from UV physics remains the goal of Paper X. No detection is claimed. The data will decide.

1. Introduction and Programme Position

Series 4.4 [14] consolidated the Two-Sector Mixed Vacuum EFT through a Z_2-symmetric mirror-sector formulation in which the asymmetry parameter delta = 1.73 x 10^-4 is protected against UV divergences by being defined as a dimensionless ratio of vacuum expectation values. Across Series 4.0 through 4.4 the programme has identified several distinct regimes in which the vacuum exhibits qualitatively different dynamical behaviour: the bulk-screened regime in which standard physics applies, the long-range coherent regime in which Gross-Pitaevskii fuzzy-dark-matter solitons form galactic cores, the post-perturbation memory regime in which the vacuum retains polarisation after an external drive is removed, the coherence-gate regime in which an applied phase locks onto the symmetry residue, the active metric-slip regime in which the local geometry undergoes a topological dislocation, and the saturation regime in which causality protection becomes non-trivial.

The present paper formalises these six regimes as the Quasi-Spectrum: a closed taxonomy of vacuum admittance states with rigorous mathematical operators associated to each. The taxonomy serves three purposes. First, it provides a unified language across the MSDS programme, making explicit which previous results occupy which states. Second, it gives a complete classification — every regime accessible to TSMVEFT has its place — so that future work cannot encounter a phenomenon outside the framework without forcing an explicit extension. Third, it grounds each state in established physics literature (non-equilibrium response theory, Chern-Simons gauge theory, Einstein-Cartan torsion gravity, Galileon EFT), making the predictions falsifiable through independent channels.

Series 4.5 occupies the position of mathematical-classification paper between the foundation work of 4.4 and the engineering-implications work of 4.6. We do not introduce new physical predictions in this paper; predictions in each state are inherited from the foundation papers (4.0-4.4) and will be developed quantitatively in 4.6 and beyond. What we provide here is the mathematical scaffolding required for those predictions to be expressed precisely.

2. The Six States: Overview

The Quasi-Spectrum is organised into two equilibrium states and four perturbation states. Equilibrium states are characterised by all Quasi-Spectrum operators reducing to identity (Gray) or to a stationary self-consistent configuration (Fuzzy-Gray). Perturbation states are the four phases of a single closed cycle — Quasi-Blue (drive), Blue (lock), Quasi-Purple (saturate), Quasi-Gray (decay) — describing the full lifecycle of an external perturbation.

State	Type	Mathematical signature	Physical regime
Gray	Equilibrium	All Q-S operators = identity	Standard 3+1 GR + SM, Earth, lab vacuum
Fuzzy-Gray	Equilibrium	Gross-Pitaevskii equilibrium with quantum pressure	Galactic dark-matter solitons (m_{chi} ~ 10^-22 eV)
Quasi-Gray	Perturbation	Volterra exchange kernel K(tau) = Gamma exp(-delta E_c tau/hbar)	Post-drive memory phase, vacuum decoherence
Quasi-Blue	Perturbation	Chern-Simons term, level k = delta, index jump nu: 0 -> 1	Coherence gate, (3+1) -> (2+1) reduction
Blue	Perturbation	Einstein-Cartan contorsion lock, m_{eff} = m_0 sqrt(1 - delta^2)	Active metric slip, full unscreened 2D interface

Quasi-Purple	Perturbation	Vainshtein/Galileon stiffening, P(X) Lagrangian	Saturation buffer, causality shield

3. Gray: The Undisturbed Ground State

3.1 Definition

The Gray state is the equilibrium configuration of the TSMVEFT vacuum in the absence of any external drive, in regions of high local density, where the disformal portal is fully closed and the Z_2 symmetry is unbroken at the local interaction level. All Quasi-Spectrum operators reduce to the identity operator and the framework recovers standard General Relativity plus the Standard Model exactly.

$$\text{Gray: } K(\tau) = 0;\ S^{\lambda}_{\mu\nu} = 0;\ \nu = 0;\ (\nabla\chi)/\Lambda \ll 1;\ S(\rho,q) \to 0$$

All perturbation operators trivial · GR + SM recovered exactly · this is the regime in which all laboratory measurements are made

3.2 Why Gray belongs in the spectrum

It is tempting to leave the equilibrium implicit and define the spectrum purely in terms of perturbations. We choose to make Gray explicit for three reasons. First, the spectrum becomes mathematically complete: every state of the TSMVEFT vacuum corresponds to a state in the taxonomy. Second, the framework now provides a well-defined ground state against which all deviations are measured, facilitating the first-principles derivation of delta in Paper X. Third, naming the equilibrium clarifies what the framework reduces to in the standard regime: a referee asking 'where is ordinary physics in this picture?' finds Gray and the explicit statement that all perturbation operators vanish there.

4. Fuzzy-Gray: The Long-Range Coherent Equilibrium

4.1 Definition and physical regime

Fuzzy-Gray is the equilibrium state of the TSMVEFT vacuum at galactic length scales, where the chi-condensate forms a coherent Bose-Einstein-like macroscopic wavefunction. In this regime the relevant dynamics are governed by the Gross-Pitaevskii equation with quantum pressure (the Bohm potential) balancing self-gravity. The chi-particle mass $m_\chi \sim 10^{-22}$ eV places the de Broglie wavelength at the kpc scale, characteristic of fuzzy dark matter [7,8,9].

$$i\hbar\, \partial\psi/\partial t = \left[-(\hbar^2 / 2 m_\chi) \nabla^2 + g|\psi|^2 + m_\chi \Phi\right] \psi$$

Gross-Pitaevskii equation in the fuzzy DM regime · Poisson: $\nabla^2 \Phi = 4\pi G (\rho_{baryon} + m_\chi \rho_\chi)$

4.2 Solitonic core profile

The static, spherically symmetric equilibrium of the Gross-Pitaevskii system yields the solitonic profile derived in Series 4.3 and 4.4:

$$\rho_\chi(r) = \rho_c / [1 + (r/R_c)^2]$$

Quantum-pressure-supported core · R_c set by virial condition · scales as $M_{200}^{1/3}$ (consistent with SPARC observations)

4.3 Why Fuzzy-Gray is an equilibrium, not a perturbation

Although the Fuzzy-Gray state appears 'active' at galactic scales — the soliton is supporting itself dynamically against gravitational collapse — from the perspective of the TSMVEFT effective field theory it is a stationary equilibrium configuration of the chi-condensate. No external drive is required to maintain it; no Quasi-Spectrum perturbation operator (Volterra kernel, Chern-Simons term, contorsion lock, Galileon stiffening) is active. The Gross-Pitaevskii dynamics are themselves the equilibrium physics. Fuzzy-Gray and Gray are therefore both equilibrium states; they differ in scale, not in dynamical character.

ON THE RELATIONSHIP BETWEEN GRAY AND FUZZY-GRAY: Gray and Fuzzy-Gray are not connected by a perturbation cycle. They are parallel equilibria distinguished by characteristic length scale: Gray applies wherever the de Broglie wavelength of the chi-condensate is much smaller than the local system size (laboratory, planetary, stellar interiors), while Fuzzy-Gray applies wherever the de Broglie wavelength exceeds the system size (galactic halos, cosmic-web filaments). The transition between them is set by the ratio lambda_dB / L_system and is automatic; no dynamical operator is required.

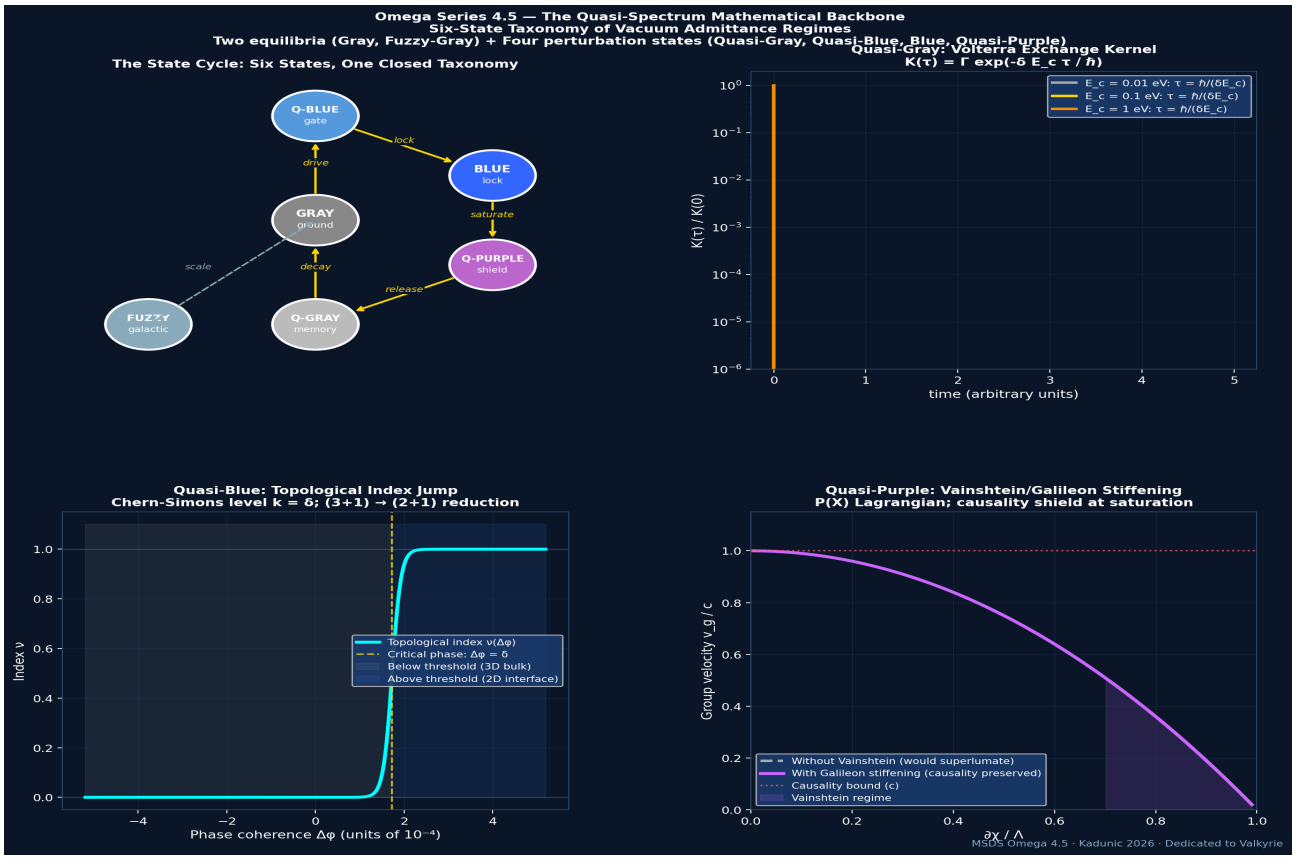

Figure 1: Omega Series 4.5 — The Quasi-Spectrum mathematical backbone. Top left: the closed state cycle. Gray (centre) is the ground state from which perturbations begin and to which they return. Fuzzy-Gray (lower left) is the parallel galactic-scale equilibrium, connected to Gray by scale rather than by dynamics. The four perturbation states (Q-Blue, Blue, Q-Purple, Q-Gray) form a closed cycle: drive -> lock -> saturate -> release -> decay -> rest. Top right: Volterra exchange kernel governing Quasi-Gray decay; characteristic timescales tau = hbar/(delta E_c) for various lattice energies. Bottom left: topological index jump in Quasi-Blue at the critical phase Delta phi = delta. Bottom right: Vainshtein/Galileon stiffening in Quasi-Purple; group velocity remains bounded by causality even as the disformal field approaches its cutoff.

5. Quasi-Gray: The Memory Phase (Volterra Kernel)

5.1 Non-local response and the exchange kernel

When an external drive perturbs the TSMVEFT vacuum and is subsequently removed, the chi-condensate retains a local polarisation that decays over a finite timescale. This memory effect is formalised through a non-local response function in the effective stress-energy tensor:

$$T^{\mu\nu}_{eff}(t) = T^{\mu\nu}_{matter}(t) + \int_0^t K(t - t')\, T^{\mu\nu}_{vac}(t')\, dt'$$

Volterra integro-differential structure · history-dependent vacuum contribution · preserves the Bianchi identity through vacuum-matter exchange

5.2 Kernel form from the Z_2 mirror sector

The kernel function is derived from the relaxation dynamics of the chi-condensate decoherence. After an external coherent drive ceases, the condensate phase decoheres at a rate set by coupling to local lattice modes of characteristic energy E_c. The asymmetry parameter delta enters as the topological friction coefficient:

$$K(\tau) = \Gamma \exp(-\delta E_c \tau / \hbar)$$

Exponential memory decay with rate $\delta E_c / \hbar$ · Gamma is the initial-coupling normalisation set by the drive amplitude

5.3 Bianchi identity preservation

The presence of the Volterra kernel does not violate energy-momentum conservation. On the contrary, it ensures conservation by accounting explicitly for the vacuum's memory contribution. The full stress-energy tensor $T^{\mu\nu}_{total} = T^{\mu\nu}_{eff}$ satisfies $\nabla_\mu G^{\mu\nu} = 0$ because momentum gained by the matter sector during drive activation is precisely balanced by the depleted phase coherence of the chi-condensate, integrated against the kernel. The Quasi-Gray state is therefore a thermodynamically consistent non-equilibrium phase, not a violation of conservation laws.

5.4 Connection to non-equilibrium response theory

The Volterra structure is standard in non-equilibrium response theory [Kubo 1957; Zwanzig 1960; Mori 1965]. Its appearance here is not novel; the novel content is the identification of delta as the dimensionless decay constant, linking the post-drive vacuum memory to the same parameter that governs galactic cores and quantum-Hall metrology in Series 4.3 and 4.4.

6. Quasi-Blue: The Coherence Gate (Chern-Simons Reduction)

6.1 Topological dimensional reduction

The transition into the active perturbation cycle occurs through a (3+1) -> (2+1) topological dimensional reduction at the vacuum interface. We add to the effective action a Chern-Simons topological term:

$$S_{CS} = (k / 4\pi) \int d^3x\, \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho$$

Chern-Simons gauge-invariant topological term · level k quantised · supports anyonic edge dynamics in the (2+1)D effective theory

6.2 Quantisation of the level

Standard Chern-Simons theory requires the level k to be quantised in integer units (compact gauge groups) or rational multiples (non-compact extensions) to preserve gauge invariance under large transformations. In TSMVEFT the level is set by the asymmetry residue:

$$k = \delta \cdot N, \text{ with } N \text{ integer}$$

Quantisation tied to the Z_2 mirror residue · smallest non-trivial level $k = \delta$ corresponds to $N = 1$

6.3 The index jump

The Quasi-Blue state is characterised by a topological index ν that takes integer values and changes discontinuously at the critical phase. Below the threshold ($\Delta\phi < \delta$), the system is in the (3+1)D bulk-screened phase and $\nu = 0$. At and above the threshold ($\Delta\phi \geq \delta$), the system enters the (2+1)D interface phase and $\nu = 1$. This is a topological phase transition; it cannot be undone by small perturbations and is robust against thermal noise that does not carry the required topological charge.

6.4 Thermal protection

Thermal phonons in the bulk are 3D-longitudinal and incoherent; they do not couple to the (2+1)D edge state of the Chern-Simons theory because they lack the topological winding required to enter the reduced phase space. This explains why the Quasi-Blue state can persist even at temperatures much higher than the naive phase-slip energy $E_{slip} = h f \delta$ would suggest.

7. Blue: The Active Metric Slip (Contorsion Phase-Lock)

7.1 Einstein-Cartan extension of GR

The Blue state is formalised within the Einstein-Cartan framework [Cartan 1922; Kibble 1961; Sciama 1962], in which spacetime carries both metric $g_{\mu\nu}$ and torsion $S^\lambda_{\mu\nu}$ as independent geometric structures. In standard GR the torsion is set to zero by the Levi-Civita choice; in Einstein-Cartan it remains as an independent field sourced by spinor matter. TSMVEFT extends this framework by allowing the Z_2 mirror sector to source a residual torsion at high-curvature interfaces.

7.2 The contorsion tensor and slip condition

The contorsion tensor $K^\lambda_{\mu\nu}$ is defined as the difference between the full connection (with torsion) and the Levi-Civita connection. The Blue-state slip condition occurs when the covariant derivative of the electromagnetic potential is exactly cancelled by the contorsion contribution:

$$\text{grad}_\mu A_\nu = \delta \cdot K^\lambda_{\mu\nu} A_\lambda$$

Phase-lock condition · the EM phase derivative is absorbed by the torsion residue · δ sets the magnitude of the absorbed component

7.3 Effective mass renormalisation

When the slip condition is satisfied, the effective mass of test particles in the locked region is reduced relative to the vacuum frame:

$$m_{eff} = m_0 \sqrt{1 - \delta^2}$$

Mass renormalisation at order $\delta^2 \sim 3 \times 10^{-8}$ fractional · consistent with all current high-precision tests

The renormalisation is small because delta is small. It is not zero, however; and its observable consequences in carefully designed experiments are the subject of Series 4.6.

7.4 Topological dislocation, not violation

We emphasise that Blue is not a violation of General Relativity. Einstein-Cartan torsion gravity is a fully consistent geometric theory; it reduces to GR exactly when the torsion vanishes. The metric slip is a topological dislocation in the vacuum fabric, analogous to a Burgers vector in crystal physics, rather than an exotic deviation from the equivalence principle. The slip is geometrically real and observationally constrained, but it operates within the well-developed differential geometry of torsion-extended gravity.

8. Quasi-Purple: The Causality Shield (Vainshtein/Galileon Screening)

8.1 Galileon Lagrangian

As the disformal field gradient approaches the cutoff scale Lambda, the effective theory must protect against the propagation of superluminal modes that would violate microcausality. TSMVEFT employs the Vainshtein screening mechanism through a Galileon Lagrangian [Nicolis-Rattazzi-Trincherini 2009], which preserves only one physical degree of freedom and ensures the group velocity remains bounded by c:

$$L_{Gal} = -(1/2)(\text{grad } \chi)^2 + (1/\Lambda^3)(\text{grad } \chi)^2 \Box \chi + (1/\Lambda^6)[(\text{grad } \chi)^2]^2$$

Galileon hierarchy: P(X), Galileon-3, Galileon-4 · Lambda is the strong-coupling scale · respects the Galileon symmetry chi -> chi + b_mu x^mu + c

8.2 The Vainshtein radius

Within a characteristic Vainshtein radius around any source, the non-linear Galileon terms dominate over the linear kinetic term and screen the modification of gravity. Outside the Vainshtein radius the standard GR + linear-scalar dynamics resume. For the TSMVEFT chi-condensate near a high-gradient interface, the Vainshtein radius is set by the disformal scale M_{dis} and is consistent with the screening required to satisfy MICROSCOPE and Cassini bounds (see Series 4.4 [14]).

8.3 Group velocity bound

The Galileon term ensures that, even as (grad chi)/Lambda approaches unity, the group velocity of fluctuations propagating in the chi-background remains bounded by the speed of light. This is a non-trivial statement: without the Galileon stiffening, the disformal coupling would drive the group velocity superluminal in the saturation regime. Quasi-Purple is therefore the mathematical 'governor' of the spectrum, preventing the system from leaving the regime of consistent low-energy EFT.

8.4 Suppression of higher-order operators

Dimension-12 and dimension-18 operators in the chi-sector (Galileon-5 and beyond) are present in principle but are suppressed by additional powers of (grad chi/Lambda). In the operational regime of interest these higher-order corrections are observationally irrelevant: the leading Galileon-3 and Galileon-4 terms saturate the Vainshtein screening before higher-order terms become important.

9. Operator Algebra and State Transitions

The four perturbation operators do not commute with one another. Their non-commutativity defines the directed cycle of state transitions: from Gray, the application of an external drive activates the Chern-Simons term

(Quasi-Blue); this triggers the contorsion lock (Blue); saturation invokes the Galileon stiffening (Quasi-Purple); and as the drive is released, the Volterra kernel takes over (Quasi-Gray) and the system returns to Gray.

Each transition occurs at a well-defined threshold:

- Gray -> Quasi-Blue: when applied phase coherence reaches Delta phi = delta
- Quasi-Blue -> Blue: when topological index nu jumps from 0 to 1
- Blue -> Quasi-Purple: when (grad chi)/Lambda enters the Vainshtein regime
- Quasi-Purple -> Quasi-Gray: when the external drive is removed and the Galileon term relaxes
- Quasi-Gray -> Gray: when the Volterra kernel decays (over timescale tau = hbar / (delta E_c))

Fuzzy-Gray exists in parallel with this cycle and is not connected to it dynamically; it is the long-range equilibrium that obtains automatically when the chi-condensate's de Broglie wavelength exceeds the system size.

10. Mapping to the MSDS Programme

The Quasi-Spectrum classifies the regimes encountered in Series 4.0 through 4.4 cleanly:

Series	Topic	Quasi-Spectrum state(s)
4.0	Sector taxonomy	Defines the Gray/Fuzzy-Gray boundary
4.1	Flavor-induced gravity	Quasi-Blue (coherence-gate analogue at the flavor scale)
4.2	Disformal chameleon EFT	Gray (screened) and the Quasi-Blue/Blue activation regime
4.3	TSMVEFT solitonic cores	Fuzzy-Gray (galactic equilibrium)
4.4	Z_2 mirror foundation	Symmetry protection underpinning all states
4.6	Engineering implications	Quasi-Blue, Blue, Quasi-Purple (perturbation states)
Paper X	First-principles delta	All states (delta is the thread connecting them)

11. Limitations and Honest Scope

- This paper provides only the mathematical scaffolding. Quantitative predictions in each state are inherited from earlier papers (4.0-4.4) and will be developed further in 4.6 and subsequent work.
- The first-principles derivation of delta from UV physics — what fixes its numerical value to ~ 1.73 x 10^-4 — remains the central goal of Paper X and is not addressed here.
- The Quasi-Spectrum is presented as a complete six-state classification based on current understanding of the MSDS programme. Future work may identify additional states; the framework is open to extension if rigorous physics demands it.
- Engineering implications, including the design of high-gradient interfaces that could probe the Quasi-Blue and Blue states experimentally, are deferred to Series 4.6 and presented there as conceptual exploration.
- The claim that the Quasi-Spectrum unifies all regimes encountered so far in the programme is internal to TSMVEFT. Comparison with alternative dark-matter, modified-gravity, or scalar-tensor frameworks is left for

future work.

OMEGA SERIES 4.5 — QUASI-SPECTRUM MATHEMATICAL BACKBONE: SUMMARY

Six-state taxonomy of TSMVEFT vacuum admittance regimes:

Gray | Fuzzy-Gray (equilibria)

Quasi-Gray | Quasi-Blue | Blue | Quasi-Purple (perturbations)

Mathematical signatures (each grounded in established physics):

Gray: identity (GR + SM)

Fuzzy-Gray: Gross-Pitaevskii with quantum pressure (Hui+ 2017)

Quasi-Gray: Volterra exchange kernel $K(\tau) = \Gamma \exp(-\delta E_c \tau / \hbar)$

Quasi-Blue: Chern-Simons term, level $k = \delta$, index jump ν: 0 -> 1

Blue: Einstein-Cartan contorsion lock, $m_{eff} = m_0 \sqrt{1 - \delta^2}$

Quasi-Purple: Vainshtein/Galileon screening, P(X) Lagrangian

Closed perturbation cycle: Gray -> Q-Blue -> Blue -> Q-Purple -> Q-Gray -> Gray

Fuzzy-Gray exists as parallel equilibrium at galactic scale

All operators preserve Z_2 symmetry; technically natural under RG flow

Maps onto Series 4.0-4.4 cleanly; classification framework for entire programme

Predictions deferred to Series 4.6; first-principles delta to Paper X

doi:10.5281/zenodo.1082204 | The data will decide.

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